

Observation Laws, Universal Quotients, and Saturated Observable Geometry

A Factorization-Theoretic Foundation for Resolution Geometry

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Abstract

We formulate observation as a process-level map $\mathfrak{F} : X_{\text{lat}} \rightarrow X_{\text{obs}}$ from latent realizations to observable process laws. Observational equivalence is equality of observable records. The quotient $Q_{\text{obs}} = X_{\text{lat}} / \sim_{\text{obs}}$ is universal for maps that are constant on observational fibres: every observationally invariant construction factors uniquely through the quotient. This elementary but exact universal property is taken as the foundational observation principle of Resolution Geometry.

A second-order shadow sends an observable law to covariance–coupling data (Σ_R, Σ_N, C) . For the orthogonal gauge action $O(p) \times O(q)$, a dimension audit gives generic quotient dimension $p+q+pq$. Consequently, the previously proposed coordinates consisting only of the two covariance spectra and the $\min(p, q)$ canonical correlations are generically incomplete. The missing continuous data record the orientation of the coupling relative to the covariance eigenspaces. On the regular stratum, the finite mixed-moment family

$$\text{tr } \Sigma_R^k, \quad \text{tr } \Sigma_N^\ell, \quad \text{tr}(\Sigma_R^a C \Sigma_N^b C^\top)$$

with appropriate finite exponent ranges has $p + q + pq$ generically independent differentials. It therefore supplies local coordinates on the continuous quotient and leaves at most finite, discrete ambiguity. We call this *generic differential saturation*; we do not claim that these quadratic mixed moments globally separate every orbit on every stratum.

Hidden-extension, higher-moment, and memory stress tests clarify the scope of the result. The covariance quotient is the complete generic continuous second-order shadow of the observation law, but it is not the full observation law and does not reconstruct latent ontology. The resulting layered architecture is

$$\begin{aligned} &\text{latent realizations} \longrightarrow \text{observable quotient} \longrightarrow \text{observable shadows} \longrightarrow \\ &\longrightarrow \text{information, coherence, entropy, and transport constructions.} \end{aligned}$$

The analytic theorems do not depend on numerical experiments; deterministic batteries are retained only as reproducibility and falsification witnesses.

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1 Scope and standing conventions

The purpose of this paper is narrow. It identifies the mathematically forced object associated with an observation law, proves its universal factorization property, and determines what can and cannot be retained by a generic second-order reduction. It does not identify observation with covariance, entropy, Fisher information, or any particular physical clock.

The sets below may be replaced by measurable spaces, topological spaces, smooth spaces, or objects in a regular category, provided the indicated quotients and morphisms exist in that category. The set-theoretic formulation is sufficient for the universal theorem and avoids adding regularity not used in its proof.

Definition 1.1 (Observation law). An observation law is a map

$$\mathfrak{F} : X_{\text{lat}} \longrightarrow X_{\text{obs}},$$

where X_{lat} is a class of latent realizations and X_{obs} is a class of observable process laws or complete observable records under a fixed protocol. The protocol is part of the definition of $\mathfrak{F}$.

Definition 1.2 (Observational equivalence). For $x, x' \in X_{\text{lat}}$, define

$$x \sim_{\text{obs}} x' \iff \mathfrak{F}(x) = \mathfrak{F}(x').$$

Let $\pi : X_{\text{lat}} \rightarrow Q_{\text{obs}} := X_{\text{lat}} / \sim_{\text{obs}}$ denote the canonical projection.

Remark 1.3 (Exact versus approximate comparison). Exact equality of observable laws defines an equivalence relation. A tolerance rule such as $d(\mathfrak{F}(x), \mathfrak{F}(x')) \leq \varepsilon$ is generally not transitive and therefore need not define a quotient. Approximate observation requires an additional construction (for example, a generated equivalence relation, a cover, a clustering rule, or a statistical decision structure); it is not silently substituted for the exact quotient used here.

2 The universal observation quotient

Theorem 2.1 (Observation quotient). *The relation $\sim_{\text{obs}}$ is an equivalence relation. Hence the set quotient*

$$Q_{\text{obs}} = X_{\text{lat}} / \sim_{\text{obs}}$$

exists, with canonical surjection $\pi : X_{\text{lat}} \rightarrow Q_{\text{obs}}$.

Proof. Reflexivity and symmetry follow from equality in X_{obs} . If $\mathfrak{F}(x) = \mathfrak{F}(x')$ and $\mathfrak{F}(x') = \mathfrak{F}(x'')$, transitivity of equality gives $\mathfrak{F}(x) = \mathfrak{F}(x'')$. Thus the equivalence classes form the stated quotient. $\square$

Theorem 2.2 (Universal observable factorization). *Let $A : X_{\text{lat}} \rightarrow Y$ be a map satisfying*

$$\mathfrak{F}(x) = \mathfrak{F}(x') \implies A(x) = A(x').$$

Then there exists a unique map $\bar{A} : Q_{\text{obs}} \rightarrow Y$ such that

$$A = \bar{A} \circ \pi.$$

Conversely, every map of the form $\bar{A} \circ \pi$ is observationally invariant.

Proof. Define $\bar{A}([x]) = A(x)$. The invariance hypothesis makes this definition independent of the representative. Then $(\bar{A} \circ \pi)(x) = \bar{A}([x]) = A(x)$. If $B \circ \pi = A$, surjectivity of π gives $B([x]) = A(x) = \bar{A}([x])$ for every class, so $B = \bar{A}$. The converse is immediate because $\pi(x) = \pi(x')$ whenever $x \sim_{\text{obs}} x'$. $\square$

Corollary 2.3 (Canonical image realization). *There is a unique bijection*

$$\tilde{\mathfrak{F}} : Q_{\text{obs}} \longrightarrow \text{im}(\mathfrak{F}), \quad \tilde{\mathfrak{F}}([x]) = \mathfrak{F}(x),$$

such that $\mathfrak{F} = \iota \circ \tilde{\mathfrak{F}} \circ \pi$, where $\iota : \text{im}(\mathfrak{F}) \hookrightarrow X_{\text{obs}}$ is inclusion.

Proof. Well-definedness and injectivity are exactly the definition of $\sim_{\text{obs}}$; surjectivity holds by the definition of the image. The factorization identity is pointwise. $\square$

Remark 2.4 (Direction of the arrows). The quotient projection is $\pi : X_{\text{lat}} \rightarrow Q_{\text{obs}}$, not a map $X_{\text{obs}} \rightarrow Q_{\text{obs}}$. The quotient is canonically identifiable with $\text{im}(\mathfrak{F})$, but not with all of X_{obs} unless $\mathfrak{F}$ is surjective. This minor typing distinction is essential for the architecture to commute.

Corollary 2.5 (Derived observable constructions). *If $D : X_{\text{obs}} \rightarrow Y$ is any construction on observable laws, then the latent map $D \circ \mathfrak{F}$ factors uniquely through π :*

$$D \circ \mathfrak{F} = \bar{D} \circ \pi, \quad \bar{D}([x]) = D(\mathfrak{F}(x)).$$

This applies, when defined, to covariance shadows, prediction operators, entropies, Fisher metrics, coherence quantities, and admissible transport data.

3 Second-order observable shadows

Definition 3.1 (Second-order shadow). Let $R \cong \mathbb{R}^p$ and $N \cong \mathbb{R}^q$. A second-order shadow is a map

$$S_2 : X_{\text{obs}} \longrightarrow \mathcal{M}_2, \quad S_2(P) = (\Sigma_R(P), \Sigma_N(P), C(P)),$$

where $\Sigma_R \in \text{Sym}_p^+$, $\Sigma_N \in \text{Sym}_q^+$, and $C \in \mathbb{R}^{p \times q}$. The latent second-order shadow is $S_2 \circ \mathfrak{F}$.

Theorem 3.2 (Second-order factorization). *The latent second-order shadow factors uniquely through the observation quotient:*

$$S_2 \circ \mathfrak{F} = \bar{S}_2 \circ \pi, \quad \bar{S}_2([x]) = S_2(\mathfrak{F}(x)).$$

Proof. If $x \sim_{\text{obs}} x'$, then $\mathfrak{F}(x) = \mathfrak{F}(x')$, hence $S_2(\mathfrak{F}(x)) = S_2(\mathfrak{F}(x'))$. Apply Theorem 2.2. $\square$

The natural change-of-frame group is

$$G = O(p) \times O(q),$$

acting by

$$(U, V) \cdot (\Sigma_R, \Sigma_N, C) = (U\Sigma_R U^\top, V\Sigma_N V^\top, UCV^\top).$$

We study the regular stratum on which both covariance matrices have simple spectra and the stabilizer of the complete triple is finite. Statements about singular spectra, repeated eigenvalues, zero coupling entries, or enlarged stabilizers require a separate stratified analysis.

4 Dimension audit and failure of the reduced coordinates

Proposition 4.1 (Generic quotient dimension). *On the regular stratum,*

$$\dim(\mathcal{M}_2/G) = p + q + pq.$$

Proof. The ambient dimension is

$$\frac{p(p+1)}{2} + \frac{q(q+1)}{2} + pq.$$

The group dimension is

$$\frac{p(p-1)}{2} + \frac{q(q-1)}{2}.$$

A finite generic stabilizer has dimension zero, so subtracting orbit dimension from ambient dimension gives $p + q + pq$. $\square$

Let $\lambda = (\lambda_1, \dots, \lambda_p)$ and $\mu = (\mu_1, \dots, \mu_q)$ be the covariance spectra, and let $\rho = (\rho_1, \dots, \rho_{\min(p,q)})$ be the singular values of the whitened coupling

$$K = \Sigma_R^{-1/2} C \Sigma_N^{-1/2}.$$

Theorem 4.2 (Incomplete fibre coordinates). *For $pq > \min(p, q)$, the family (λ, μ, ρ) cannot provide local coordinates on the regular quotient and cannot generically separate its continuous orbits.*

Proof. The family has at most $p + q + \min(p, q)$ independent real components, whereas Proposition 4.1 gives quotient dimension $p + q + pq$. The rank of a smooth map cannot exceed the dimension of its target. The deficit is $pq - \min(p, q) > 0$. $\square$

Theorem 4.3 (Orientation gap). *Canonical correlations determine coupling magnitudes after whitening, but do not in general determine the orientation of the coupling singular frames relative to the eigenspaces of Σ_R and Σ_N . In particular, on an open dense set the invariant*

$$I_{11} = \text{tr}(\Sigma_R C \Sigma_N C^\top)$$

is not a function of (λ, μ, ρ) alone.

Proof. Fix diagonal positive matrices Σ_R and Σ_N with simple spectra and fix a non-scalar diagonal singular-value matrix D . For orthogonal U, V , set

$$K_{U,V} = U D V^\top, \quad C_{U,V} = \Sigma_R^{1/2} K_{U,V} \Sigma_N^{1/2}.$$

All triples have the same covariance spectra and the same singular values of K . However,

$$I_{11}(U, V) = \text{tr}(\Sigma_R^{3/2} U D V^\top \Sigma_N^2 V D U^\top \Sigma_R^{1/2})$$

varies with U, V unless the relevant spectra or singular values are degenerate. Thus the relative frame orientation is additional invariant data. $\square$

5 Generic differential saturation

For $1 \leq k \leq p$, $1 \leq \ell \leq q$, and $0 \leq a \leq p-1$, $0 \leq b \leq q-1$, define

$$\begin{aligned} P_k &= \text{tr}(\Sigma_R^k), \\ Q_\ell &= \text{tr}(\Sigma_N^\ell), \\ M_{ab} &= \text{tr}(\Sigma_R^a C \Sigma_N^b C^\top). \end{aligned}$$

There are $p + q + pq$ such quantities.

Theorem 5.1 (Generic differential saturation closure). *Assume that Σ_R and Σ_N have simple spectra and, in their ordered eigenbases, every coupling entry c_{ij} is nonzero. Then*

$$\{P_k\}_{k=1}^p \cup \{Q_\ell\}_{\ell=1}^q \cup \{M_{ab}\}_{\substack{0 \leq a < p \\ 0 \leq b < q}}$$

has $p + q + pq$ linearly independent differentials transverse to the generic G -orbits. Consequently it defines local coordinates on the continuous regular quotient, up to a finite ambiguity.

Proof. Use the orthogonal gauge to diagonalize

$$\Sigma_R = \text{diag}(\lambda_1, \dots, \lambda_p), \quad \Sigma_N = \text{diag}(\mu_1, \dots, \mu_q),$$

with ordered, pairwise distinct eigenvalues. The power-sum Jacobians are Vandermonde matrices multiplied by nonzero diagonal factors:

$$\frac{\partial P_k}{\partial \lambda_i} = k \lambda_i^{k-1}, \quad \frac{\partial Q_\ell}{\partial \mu_j} = \ell \mu_j^{\ell-1}.$$

They have ranks p and q .

After fixing these spectral coordinates, write $z_{ij} = c_{ij}^2$. Then

$$M_{ab} = \sum_{i=1}^p \sum_{j=1}^q \lambda_i^a \mu_j^b z_{ij}.$$

The Jacobian with respect to the pq variables z_{ij} is the Kronecker product of two Vandermonde matrices,

$$V_\lambda \otimes V_\mu,$$

and is invertible because both spectra are simple. Since $c_{ij} \neq 0$, the change from c_{ij} to z_{ij} has nonzero diagonal derivative. Combining the spectral and mixed blocks yields total transverse rank $p + q + pq$, equal to the quotient dimension in Proposition 4.1. The residual ordered-eigenbasis stabilizer is finite, so only finite ambiguity remains. $\square$

Corollary 5.2 (No additional generic continuous second-order coordinates). *On the regular stratum, every smooth second-order gauge invariant is locally a smooth function of the saturated family in Theorem 5.1, after choosing a branch through the finite residual ambiguity.*

Proof. The saturated family is a local coordinate system on the continuous quotient, so every other smooth invariant descends locally to a function of those coordinates. $\square$

Non-claim 5.3 (Global orbit separation). The quadratic mixed moments recover c_{ij}^2 in covariance eigenbases, not all discrete sign-cycle data. Therefore Theorem 5.1 is a local, generic, differential saturation theorem. It does not claim global separation of all G -orbits. Global polynomial separation may require traces of longer alternating words in $\Sigma_R, \Sigma_N, C, C^\top$, and singular strata require their own stabilizer analysis.

6 Latent underdetermination

Theorem 6.1 (Latent underdetermination). *The observation law determines a latent realization only when $\mathfrak{F}$ is injective. In general,*

$$\mathfrak{F}(x) = \mathfrak{F}(x')$$

does not entail $x = x'$; it entails only $\pi(x) = \pi(x')$. Moreover, latent state multiplicity, latent dimension, and hidden topology are not invariants of the observation law whenever observationally silent extensions are admitted.

Proof. The first statement is the definition of injectivity. For the extension statement, let H be any nonempty hidden set and define $X'_{\text{lat}} = X_{\text{lat}} \times H$ with

$$\mathfrak{F}'(x, h) = \mathfrak{F}(x).$$

Then $\text{im}(\mathfrak{F}') = \text{im}(\mathfrak{F})$ and $(X_{\text{lat}} \times H) / \sim'_{\text{obs}} \cong Q_{\text{obs}}$, independently of the cardinality, dimension, or topology assigned to H . Hence such hidden structure cannot be recovered from the observation law alone. $\square$

Corollary 6.2. *No invariant that factors only through Q_{obs} can distinguish observationally silent latent extensions.*

7 Why the second-order shadow is not the full law

Proposition 7.1 (Higher-moment escape). *There exist distinct observable laws with identical means and covariance matrices. Therefore equality of second-order shadows does not imply equality in X_{obs} whenever X_{obs} retains higher-order distributional information.*

Proof. In one dimension, a centered Gaussian with variance one and a centered, non-Gaussian law scaled to variance one have the same first two moments but different higher moments or characteristic functions. Embedding either law as a component of a process gives identical prescribed second-order data but distinct observable laws. $\square$

Proposition 7.2 (Memory escape). *There exist processes with the same one-time marginal and equal-time covariance but different temporal laws. Hence an equal-time second-order shadow does not in general determine memory or transition structure.*

Proof. Take two stationary processes with the same centered unit-variance marginal, one independent in time and one with nonzero lag correlation. Their equal-time covariances agree, while their two-time distributions and prediction operators differ. $\square$

Remark 7.3. These are boundary witnesses, not failures of Theorem 5.1. That theorem closes only the generic continuous information contained in the specified second-order triple.

8 Observation architecture

Theorem 8.1 (Observation architecture). *The observation layer is represented by the commuting factorization*

$$\begin{array}{ccc} X_{\text{lat}} & \xrightarrow{\mathfrak{F}} & X_{\text{obs}} \\ \pi \downarrow & & \downarrow D \\ Q_{\text{obs}} & \xrightarrow{\overline{D}} & Y \end{array}$$

for every construction $D : X_{\text{obs}} \rightarrow Y$, where $\overline{D}([x]) = D(\mathfrak{F}(x))$. Equivalently,

$$\mathfrak{F} = \iota \circ \tilde{\mathfrak{F}} \circ \pi, \quad \tilde{\mathfrak{F}} : Q_{\text{obs}} \xrightarrow{\sim} \text{im}(\mathfrak{F}).$$

Thus every admissible observable construction descends uniquely to the quotient.

Proof. The first diagram is Theorem 2.2 applied to $D \circ \mathfrak{F}$; the second identity is Corollary 2.3. $\square$

The derived targets Y may include, under their respective hypotheses,

$$\left\{ \begin{array}{l} \text{covariance and coupling geometry,} \\ \text{Fisher or other information geometry,} \\ \text{coherence and duration observables,} \\ \text{admissible transport classes,} \\ \text{entropy functionals and prediction operators.} \end{array} \right.$$

The theorem asserts factorization, not the automatic existence, uniqueness, or physical interpretation of any one of these derived structures.

9 Deterministic stress batteries

The proofs above are analytic. Computation tests implementations, catches typing and rank errors, and probes theorem boundaries; it does not replace the proofs.

Attack	Outcome	Mathematical interpretation
Hidden extension	Pass	Quotient class count and quotient law remain invariant.
Reparameterization	Pass	Observable projector transforms co-variantly.
Factorization and fibre annihilation	Pass	Observable maps are constant on fibres and reconstruct from quotient data.
Strict refinement	Rejected	Splitting an observational class without new records is not observable.
Strict coarsening	Rejected	Merging distinguishable classes loses the law.
Orientation recovery	Pass on regular samples	Mixed moments recover continuous frame-relative coupling data.
Higher-moment escape	Expected boundary witness	Second-order equality does not imply equality of full laws.
Memory escape	Expected boundary witness	Equal-time covariance does not determine temporal law.

The final finite observation-law battery used seed 20260818 and passed all 16 registered checks. Its base instance had four observable classes, observable rank 4, null rank 3, and contrast rank 3. Companion fixed-protocol and duration/clock batteries use seeds 20260805 and 20260806, respectively; they are supporting studies and are not premises of any theorem in this paper.

10 Non-claims and precise closure statement

Non-claim 10.1. This paper does not claim reconstruction of latent reality, identification of a preferred latent dimension, completeness beyond the declared observation protocol, global separation of singular second-order orbits by quadratic mixed moments, derivation of a unique connection from covariance data, or reduction of observation to entropy, Fisher information, coherence, transport, or time.

Non-claim 10.2. The universal factorization theorem is a quotient universal property. Its role is foundational and organizational, not a claim that the underlying abstract result is new. The substantive second-order result is the dimension audit and the generic differential saturation theorem, both with explicit regularity and scope restrictions.

11 Conclusion

Observation begins with a law $\mathfrak{F}$ and the equivalence relation induced by its fibres. The quotient Q_{obs} is the minimal exact carrier of all observationally invariant constructions: every such construction factors uniquely through it. A covariance–coupling geometry is one derived shadow. Its reduced spectral/canonical-correlation coordinates are dimensionally incomplete, because they omit coupling orientation relative to the covariance eigenframes. Finite mixed moments repair the continuous deficit on the regular stratum and provide a saturated local second-order coordinate system. Higher moments, temporal memory, and hidden extensions delimit the result: the saturated covariance geometry is neither the full process law nor latent ontology.

Observation is not latent ontology, not covariance geometry, and not an entropy functional.

Observation is the passage from latent realizations to observational equivalence classes.

The resulting quotient is universal for every admissible observable construction.

A Reproducibility Record

The analytic results of the paper do not depend on numerical experiments. The computational batteries serve only as reproducibility witnesses, implementation checks, and boundary tests of the theorem ladder.

Three deterministic battery families were used:

1. Fixed-protocol uniqueness battery (seed 20260805).

This battery verifies that the observable quotient is uniquely determined by the observation protocol and that all observable constructions factor through the resulting quotient.

2. Duration and clock batteries (seed 20260806).

These tests verify protocol-invariant duration observables, quotient-level clock constructions, and consistency of observable-time representations under admissible refinements and reparameterizations.

3. Saturation and observation-law batteries.

These tests investigate generic orbit closure, orientation recovery, hidden extensions, higher-moment escapes, and memory escapes. The goal is to determine which structures are recoverable from the second-order observable shadow and which belong to the observation law itself.

The primary artifacts are

- `latent_observation_saturation_demo.json`,
- `latent_observation_saturation_battery.json`,
- `latent_observation_saturation.py`.

The demonstration instance records the explicit quotient construction, observable projector, observable/null decomposition, contrast projector, and associated ranks.

The final observation-law battery records the following checks:

- class count invariance,
- rank-nullity consistency,
- projector idempotence,
- projector symmetry,
- law factorization,
- fibre annihilation,
- hidden-extension class-count invariance,
- hidden-extension quotient-law invariance,
- reparameterization covariance,
- rejection of strict unobservable refinement,
- detection of information-losing coarsening,
- factorization of random observables,
- rejection of generic latent observables,
- contrast-rank bound,
- contrast reconstruction,
- non-transitivity witness for approximate observation relations.

All registered observation-law checks passed.

For the final deterministic run:

$$\text{passed} = 16, \quad \text{total} = 16.$$

The recorded observable quotient consists of four equivalence classes, with observable rank

$$\text{rank}(Q_{\text{obs}}) = 4,$$

null rank

$$\text{rank}(N_{\text{obs}}) = 3,$$

and contrast rank

$$\text{rank}(C_{\text{obs}}) = 3.$$

The recorded quotient transition law is

$$\begin{pmatrix} 0.7 & 0.2 & 0.1 & 0 \\ 0.1 & 0.7 & 0.1 & 0.1 \\ 0.25 & 0.25 & 0.25 & 0.25 \\ 0 & 0.2 & 0.3 & 0.5 \end{pmatrix}.$$

The saturation studies support the following empirical conclusions:

1. Hidden latent extensions preserve the observable quotient.
2. Observable constructions factor through quotient classes.
3. The reduced coordinate family (λ, μ, ρ) is generically incomplete.
4. Orientation-sensitive invariants close the generic continuous second-order orbit dimension.
5. Higher moments can escape second-order geometry.
6. Temporal memory can escape second-order geometry.

These numerical results are not premises of any theorem in the paper. They are reproducibility metadata supporting the implementation of the observation-law, quotient, and saturation constructions.

All theorem statements remain analytic and are proved independently of the computational batteries.

B Proof dependency ladder

1. Equality of observable laws induces an equivalence relation.
2. The equivalence relation yields the quotient and its universal property.
3. Every shadow of the observable law factors through that quotient.
4. The second-order gauge quotient has generic dimension $p + q + pq$.
5. Spectra plus canonical correlations are therefore incomplete.
6. Mixed Vandermonde moments supply full generic transverse differential rank.
7. This closes the continuous regular second-order sector, but not discrete, singular, higher-order, temporal, or latent structure.

References

- [1] C. Procesi, “The invariant theory of $n \times n$ matrices,” *Advances in Mathematics* 19 (1976), 306–381.
- [2] R. A. Horn and C. R. Johnson, *Matrix Analysis*, 2nd ed., Cambridge University Press, 2013.
- [3] S. Amari and H. Nagaoka, *Methods of Information Geometry*, AMS/Oxford University Press, 2000.
- [4] M. L. Eaton, *Group Invariance Applications in Statistics*, Institute of Mathematical Statistics, 1989.